EE2001 Electrical Circuits- I:

Continuation to:

Inductance, Capacitance and Mutual Inductance

28 November 2023

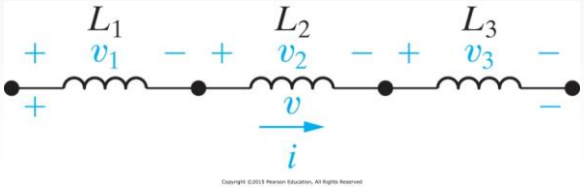
Outline



3. Series-Parallel Combinations of Inductance and Capacitance

Series-Parallel Combinations of Inductance and Capacitance

- Consider inductors are connected in series



$$v_1 = L_1 \frac{di}{dt} \quad v_2 = L_2 \frac{di}{dt} \quad v_3 = L_3 \frac{di}{dt}$$

- The same current flows through all inductors

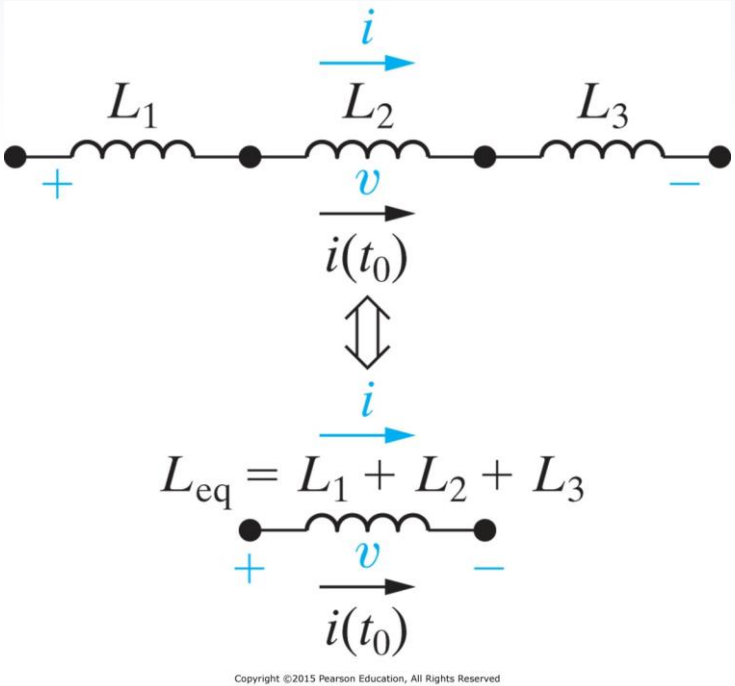
$$v = v_1 + v_2 + v_3 = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt} = (L_1 + L_2 + L_3) \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2 + L_3$$

- If the inductors carry an initial current $i(t_0)$, the equivalent inductor carries the same initial current $i(t_0)$

Series-Parallel Combinations of Inductance and Capacitance

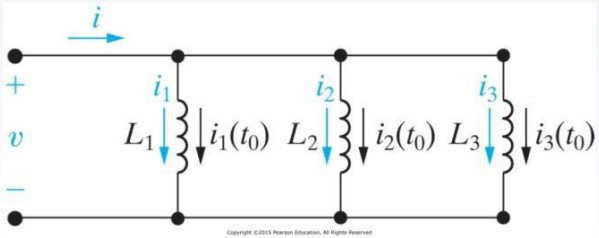
- Inductors connected in series



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Series-Parallel Combinations of Inductance and Capacitance

- Consider inductors connected in parallel



$$i_1(t) = i_1(t_0) + \frac{1}{L_1}\int_{t_o}^{t} v(\tau)\,d\tau$$

$$i_2(t) = i_2(t_0) + \frac{1}{L_2}\int_{t_o}^{t} v(\tau)\,d\tau$$

$$i_3(t) = i_3(t_0) + \frac{1}{L_3}\int_{t_o}^{t} v(\tau)\,d\tau$$

- The same voltage appears across all inductors

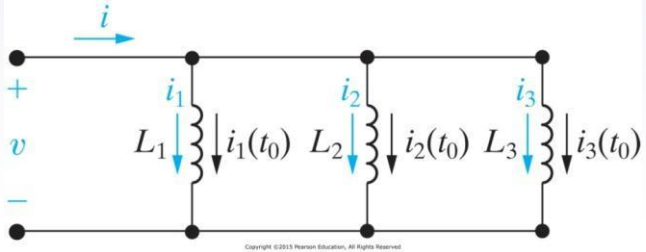
$$i = i_1 + i_2 + i_3 = \underbrace{(i_1(t_0) + i_2(t_0) + i_3(t_0))}_{i(t_0)} + \underbrace{\left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right)}_{\frac{1}{L_{eq}}} \int_{t_o}^{t} v(\tau)\,d\tau$$

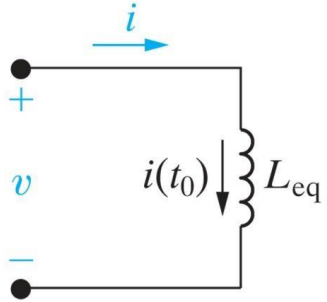
$$i = i(t_0) + \frac{1}{L_{eq}}\int_{t_o}^{t} v(\tau)\,d\tau$$

- The initial current $i(t_0)$ of the equivalent inductor is equal to the sum of the initial currents $i_1(t_0)$, $i_2(t_0)$, $i_3(t_0)$ in the individual inductors.

Series-Parallel Combinations of Inductance and Capacitance

- Inductors connected in parallel





$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$$

$$i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

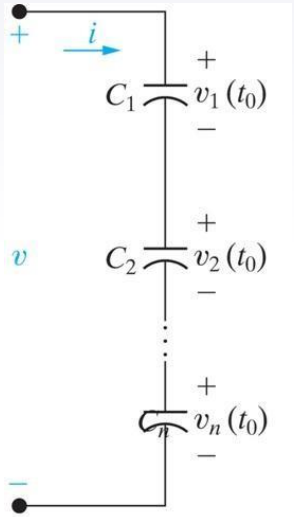
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Series-Parallel Combinations of Inductance and Capacitance

- Inductors are combined similar to resistors.
- Don’t forget the initial currents.

Series-Parallel Combinations of Inductance and Capacitance

- Consider capacitors connected in series



- The same current flows through all capacitors

$$v_1(t) = v_1(t_0) + \frac{1}{C_1}\int_{t_0}^{t} i(\tau)\,d\tau$$

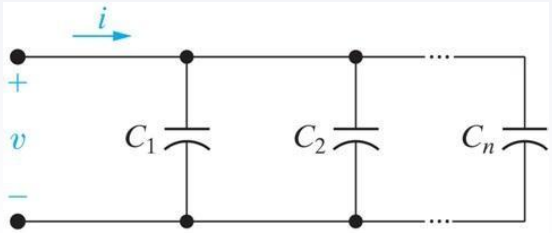
$$v_2(t) = v_2(t_0) + \frac{1}{C_2}\int_{t_0}^{t} i(\tau)\,d\tau$$

$$v_3(t) = v_3(t_0) + \frac{1}{C_3}\int_{t_0}^{t} i(\tau)\,d\tau$$

$$v = v_1 + v_2 + \cdots + v_n = \underbrace{(v_1(t_0) + v_2(t_0) + \cdots + v_n(t_0))}_{v(t_0)} + \underbrace{\left(\frac{1}{C_1} + \frac{1}{C_2} + \cdots \frac{1}{C_n}\right)}_{\frac{1}{C_{eq}}} \int_{t_0}^{t} i(\tau)\,d\tau$$

Series-Parallel Combinations of Inductance and Capacitance

- Consider capacitors connected in parallel



$$i_1 = C_1 \frac{dv}{dt}$$

$$i_2 = C_2 \frac{dv}{dt}$$

$$i_3 = C_3 \frac{dv}{dt}$$

- The same voltage appears across all capacitors

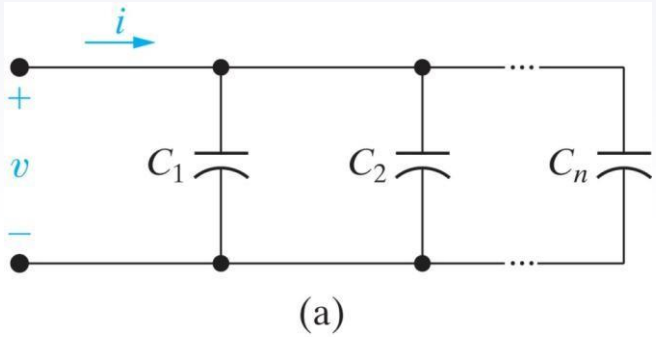
$$i = i_1 + i_2 + i_3 = \left(C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \cdots C_n \frac{dv}{dt}\right) = \underbrace{(C_1 + C_2 + \cdots + C_n)}_{C_{eq}} \frac{dv}{dt}$$

$$C_{eq} = C_1 + C_2 + \cdots + C_n$$

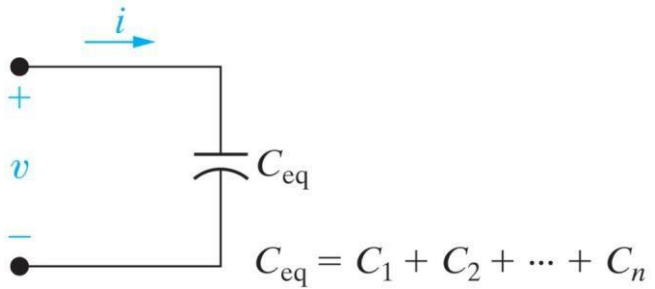
- If there is an initial voltage $v(t_0)$ over the capacitors, the same initial voltage $v(t_0)$ appears over the equivalent capacitor as well.

Series-Parallel Combinations of Inductance and Capacitance

- Capacitors connected in parallel



(a)



(b)

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Summary

TABLE 6.1 Terminal Equations for Ideal Inductors and Capacitors

Inductors	
$v = L\frac{di}{dt}$	(V)
$i = \frac{1}{L}\int_{t_0}^{t} v\,d\tau + i(t_0)$	(A)
$p = vi = Li\frac{di}{dt}$	(W)
$w = \frac{1}{2}Li^2$	(J)
Capacitors	
$v = \frac{1}{C}\int_{t_0}^{t} i\,d\tau + v(t_0)$	(V)
$i = C\frac{dv}{dt}$	(A)
$p = vi = Cv\frac{dv}{dt}$	(W)
$w = \frac{1}{2}Cv^2$	(J)

TABLE 6.2 Equations for Series- and Parallel-Connected Inductors and Capacitors

Series-Connected
$L_{eq} = L_1 + L_2 + \cdots + L_n$
$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$
Parallel-Connected
$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}$
$C_{eq} = C_1 + C_2 + \cdots + C_n$

End of the Lesson (28.11.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist